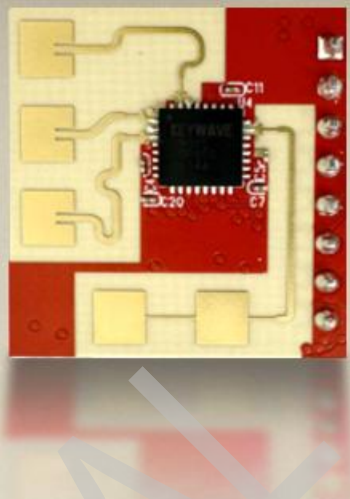
KEYWAVE

Keywave-KW307

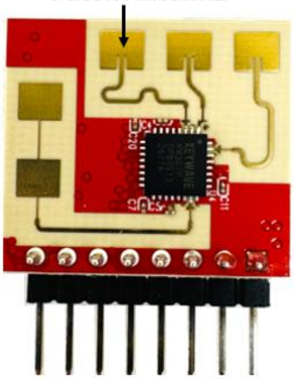
24GHz Human Presence Sensor Module

- Revolutionary new Radar technology
- Ultra-low power consumption
- Detects multiple and stationary People
- Reduces false alarms

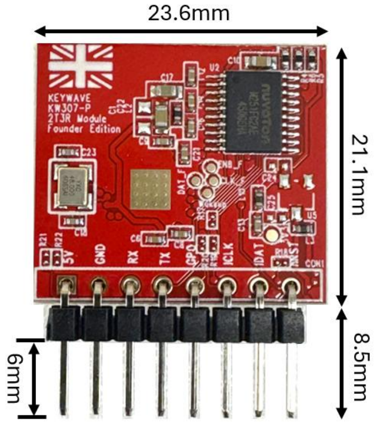


User Guide and Datasheet

Version 2.5 | 2026-08-25, Keywave Technology Ltd.



Front view
(antenna facing forward)



Back view

Side View

Product Overview

The **KW307** is a compact 24 GHz RF-based **human presence sensor module** designed for high-accuracy motion and stationary human detection in indoor and outdoor applications. Built upon a custom-designed sensing SoC and proprietary mathematical signal algorithms, the KW307 delivers real-time spatial coordinates (±5 cm distance, ±3° angle) and multi-target tracking for up to 3 individuals with ultra-low power consumption.

Key Features

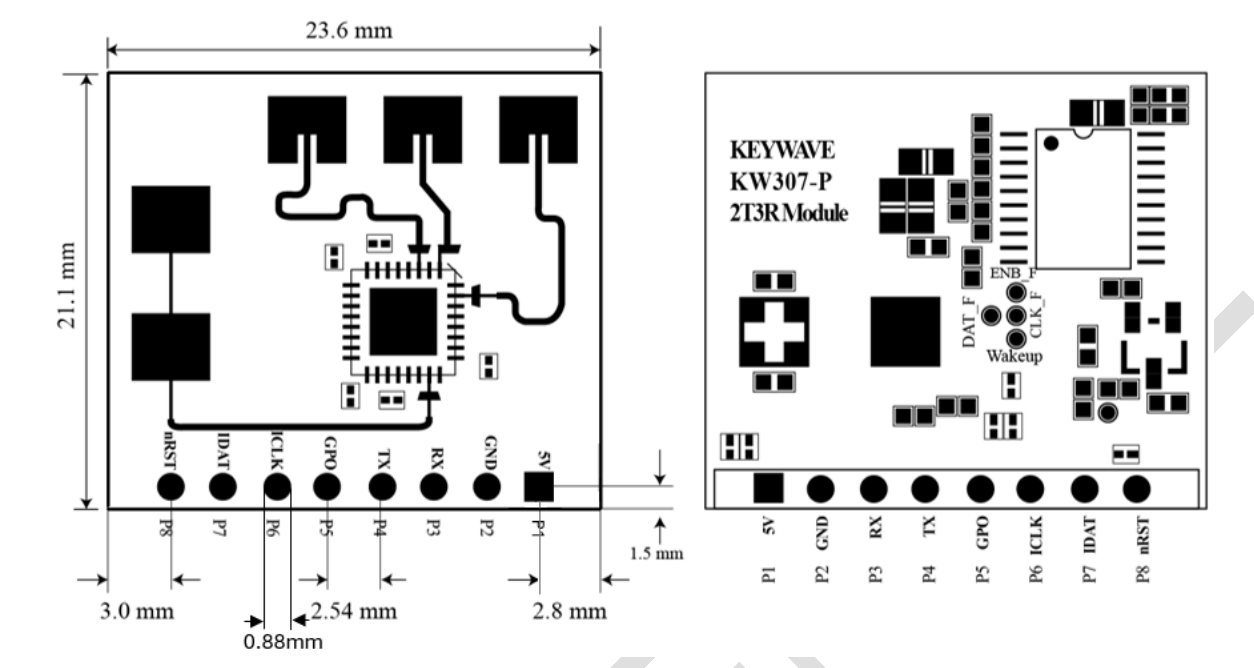
- **Proprietary Silicon Architecture:** Built from the IC level up, integrating proprietary mathematical sensing algorithms and dedicated hardware processing directly into the SoC.
- **High-Dynamic Motion & Stationary Sensing:** Detects moving and stationary humans up to 20 meters with a continuous 20Hz update rate.
- **Precise Spatial Location:** Delivers accurate real-time target distance and angle coordinates. Tracks up to 3 people simultaneously (±5 cm, ±3°).
- **Intrinsic Clutter Immunity:** Fully immune to environmental clutter, including wind, curtain, tree and ceiling fan noise.
- **Ultra-Low Power Operations:** Consumes only 12 mW at 20 Hz update rate (2 mW at 0.5 Hz).
- **Flexible Interfaces & Configuration:** Supports 3.6V–5.0V power input, UART target reporting, configurable GPO presence flag, and non-volatile memory (NVM) parameter persistence.

Application Scenarios

- Smart Building & Automation. (Lighting, HVAC, Energy Optimization)
- Space Occupancy & Desk Management Systems.
- Washroom & Industrial Automation. (Urinals, Hand Dryers)
- Security & Perimeter Presence Detection

*Recommended clearance zone: at least 8mm in total (front and back).

Module Dimensions & Pin Definitions



Description	Min.	Typ.	Max.	Unit
Module Length	**23.58**	**23.6**	**23.62**	**mm**
Module Width	**21.08**	**21.1**	**21.12**	**mm**
Module Thickness	**1.57**	**1.6**	**1.63**	**mm**
Hole Dimension	**0.875**	**0.88**	**0.9**	**mm**
Pin Header Diameter		**0.6**		**mm**

Pin	Symbol	Type	Description
P1	5V	Power	5V Power Supply Input (3.6V ~ 5.5V)
P2	GND	Power	Ground
P3	RX	Input	UART Receive Signal (3.3V Logic Level)
P4	TX	Output	UART Transmit Signal (3.3V Logic Level)
P5	GPO	Output	General Purpose Output. (Human Presence Flag output)
P6	*ICLK*	*Input*	*Programming Clock Signal.* * (Leave unconnected in typical applications)
P7	*IDAT*	*Input*	*Programming Data Signal.* * (Leave unconnected in typical applications)
P8	*nRST*	*Input*	*Active-Low System Reset* * (Leave unconnected in typical applications)

- 5V Power Mode: Supply voltage is regulated by an on-chip LDO to power the internal core.

Block Diagram

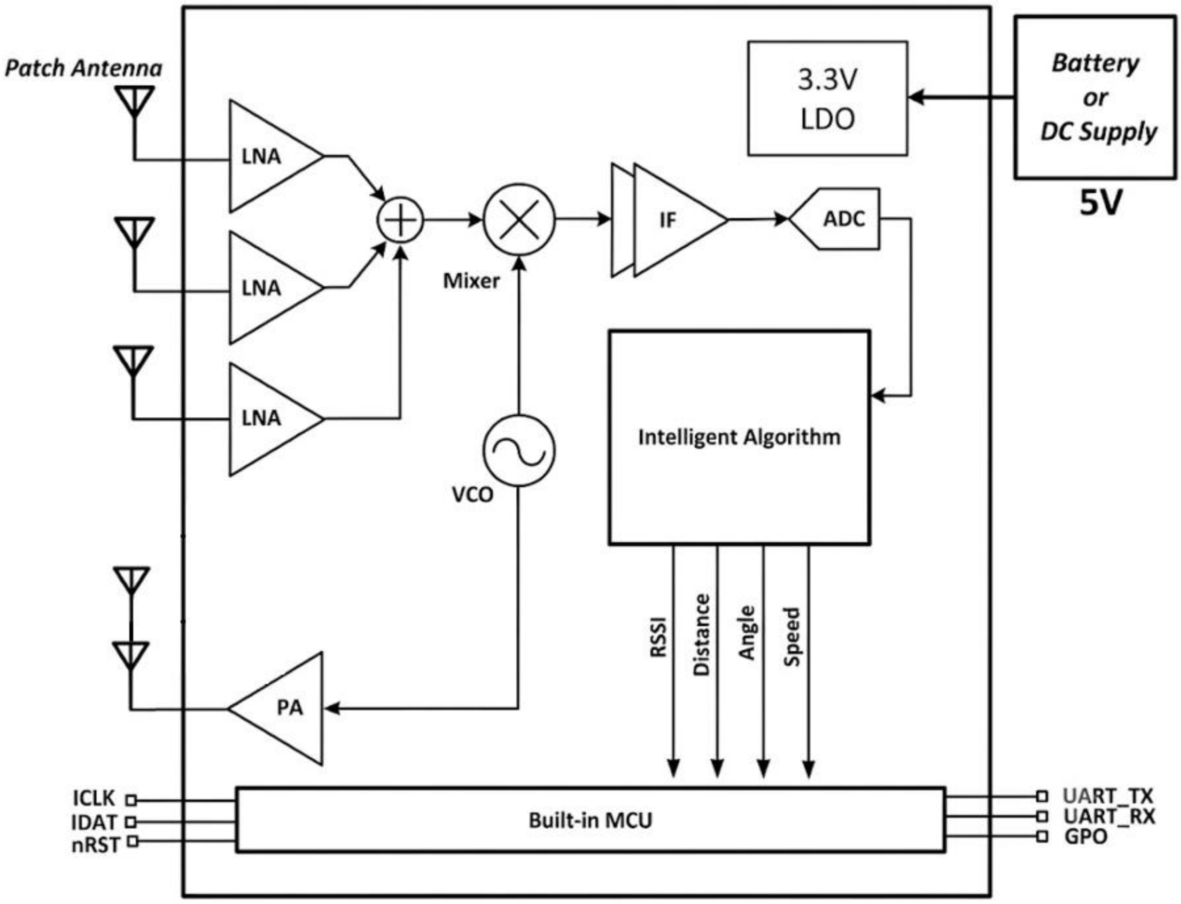


Figure : KW307 Block Diagram

Electrical Specification

Absolute Maximum Ratings

Characteristic	Symbol	Min.	Typ.	Max.	Units
5V supply voltage	V_{DD5V}	-0.3	5.0	5.5	V
Digital I/O voltage	V_{DIO}	-0.3	3.3	3.6	V
Digital I/O maximum current capability	$I_{DIO,max}$			10	mA
Ambient Temperature	T_A	-25		85	°C

* GPO cannot directly drive a high-current load; it requires a power MOS switch to do so.
* It is recommended to add a 10μF decoupling capacitor to power input.

DC Characteristics

Characteristic	Symbol	Min.	Typ.	Max.	Units
Supply voltage via 5V pin	V_{DD5V}	3.6	5.0	5.5	V
Operation Current @ 20Hz update rate	I_{DC}		4.0		mA
Operation Current @ 0.5Hz update rate	I_{DC}		0.8		mA

RF Characteristics

Characteristic	Symbol	Min.	Typ.	Max.	Units
Operation frequency	F_{RF}	24.05		24.25	GHz
RF output power	P_{RF}		-2		dBm
Detection range	Range	0.1	10	20*	m
Field of View	FOV		100		degrees

*Horizontal alignment at 1m height.
*For detection requirements up to 20 meters, please contact Keywave representative

Mounting Suggestion

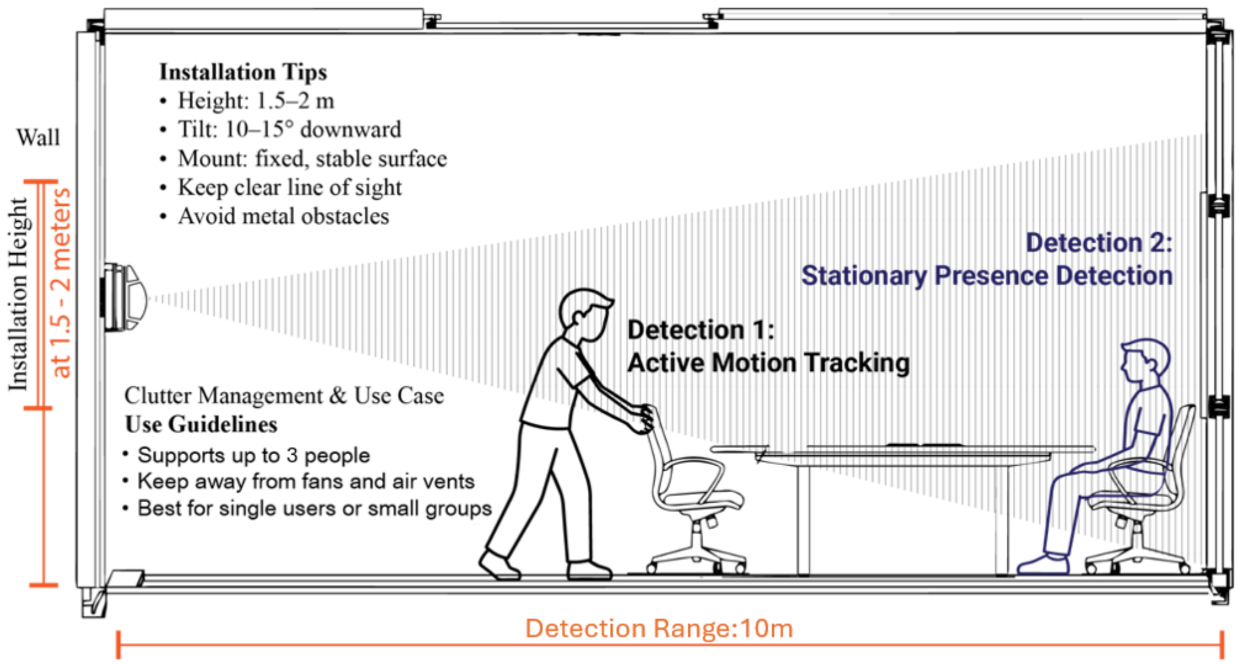
•**Optimal Installation Height**: For best results, mount the sensor at a height of 1.5m to 2m above the floor.

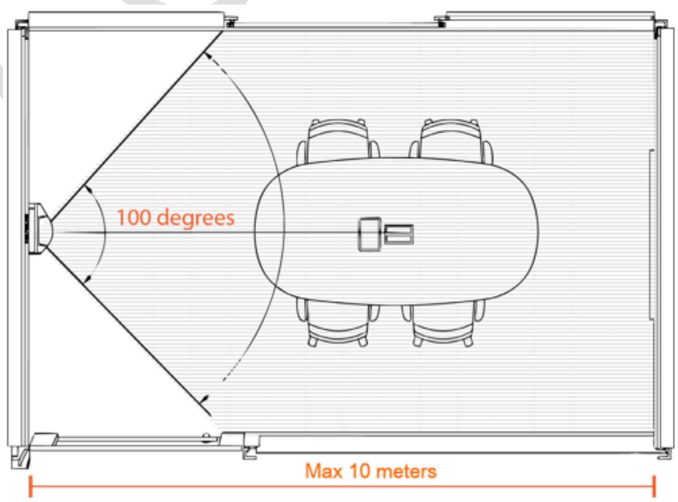
•**Avoid Obstructions**: Ensure a clear line of sight. Avoid placing the sensor behind metallic objects or thick glass, as these may block or attenuate the radar signal.

•**Directional Alignment**: If the target area is close to the mounting point tilt the sensor slightly downward (approx. 10°–15°) to maximize FOV coverage.

•**Stability**: Mount the sensor on a rigid surface. Although the KW307 has superior immunity to vibration, a stable mount ensures the highest possible accuracy for stationary detection.

•**Clutter Management:** Position the sensor away from large moving mechanical parts (such as large industrial fans) to maintain the best tracking performance for multiple people.





Configuration and monitoring tool

The KW307 configuration and monitoring tool is a GUI application designed to help you get up and running as fast as possible.

The tool allows you to:

- Configure sensor detection parameters, making it easy to tune the KW307 to your application
- Choose pre-set tuning parameters for common applications.
- Monitor real-time detection status, on an intuitive graphical display.

UART to USB Interface Board
System Connection Diagram

- The configuration tool uses a USB to UART adapter to communicate from a Windows PC to the KW307.
- Note the Pin spacing for the KW307 is 2.54mm.
- UART RX/TX interface must be 3.3V. (Not 5V Tolerant).

CAUTION :

- When connecting the adaptor, or bridge board, ensure you use a data-capable USB cable, not a charge only cable.
- Take care when connecting the cable as connection errors may permanently damage the hardware module.
- The configuration tool software can be found at: https://keywavetech.co.uk/products

GUI Configuration Overview

- Please follow the 5 steps below to complete the parameter configuration.

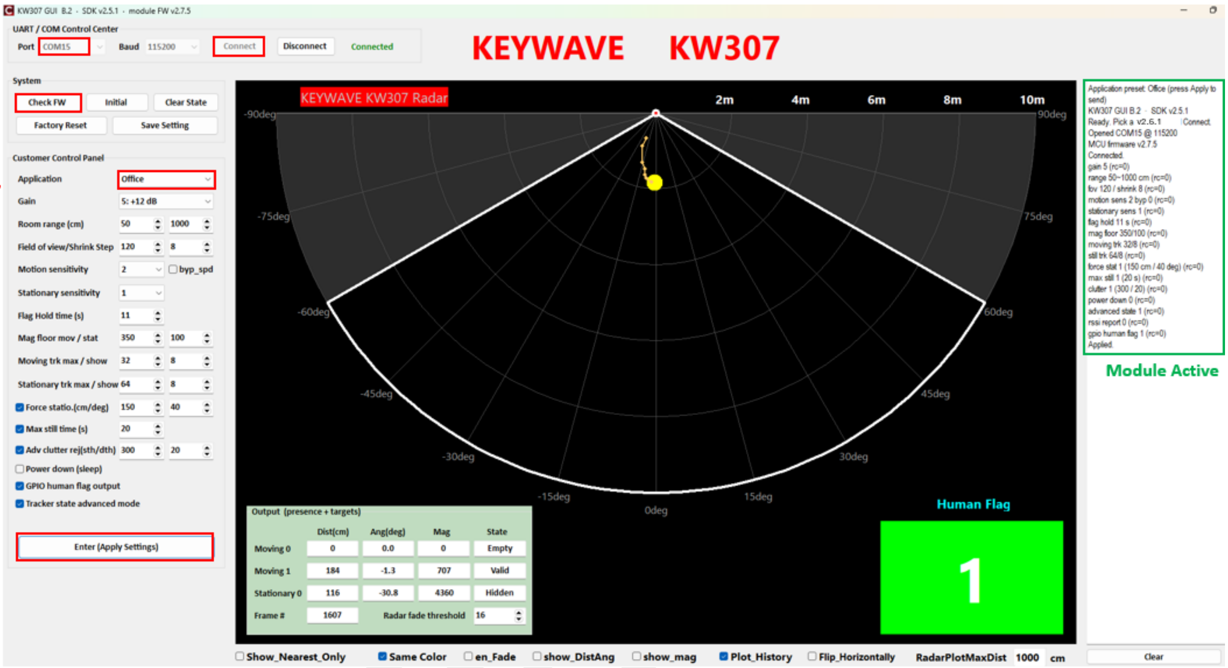
Step1 : Select "COM Port"

Step2 : Click "Connect"
(to connect com port)

Step3 : Click "Check FW"
(It should be v2.7.5)

Step4 : Select "Application"

Step5 : Click "Enter"
(Setup the setting)



- The pre-loaded application scenarios come with pre-defined optimal parameters, while still allowing users to fine-tune settings for specific applications. Please click "Enter" after you complete the desire settings.

Function	Range	Description
Application Mode	5 scenarios	Outdoor, Office, Desk, Urinal, Hand_dryer
Gain	-3db ~ +21db	Adjust radar signal strength. (+21dB: the maximum adjustable signal strength.)
Room range (min/max)	10 ~ 1200	Set the range size for radar detection. (cm)
Field of view (FOV)	0 ~ 120	Set the radar field of view. (degree)
FOV Shrink step	0 ~ 25	The FOV narrows degree per 100cm. default:8 means shrink 8deg/100cm
Motion Sensitivity	L0 ~ L9	The ability to detect a moving human. L0: Most sensitive (i.e., Wave hands or shake heads) L9: Least sensitive (i.e., Walk one step)
Stationary Sensitivity	L0 ~ L9	The ability to detect subtle movement of a stationary human. L0: Most sensitive (i.e., Breath) L9: Least sensitive (i.e., Wave hands or shake heads)
Flag Hold time	0 ~ 3000	Set the hold time of Human Flag as human leaves detection area (Sec).

Function	Range	Description
Mag floor (threshold) motion	0 ~ 2000	Signal strength above the mag threshold will be identified as valid targets for a moving human. (Environment-dependent, default: 350)
Mag floor (threshold) stationary	0 ~ 2000	Signal strength above the mag threshold will be identified as valid targets for a stationary human. (Environment-dependent, default: 100)
Motion tracker max / show	1 ~ 64	Max: The maximum number of frames a moving target survives in the tracker plot without receiving any new detection updates. Show: The detection count threshold required for a newly initialized target to transition into the "Shown" (active/confirmed) state.
Stationary tracker max / show	1 ~ 64	Max: The maximum number of frames a stationary target survives in the tracker plot without receiving any new detection updates. Show: The detection count threshold required for a newly initialized target to transition into the "Shown" (active/confirmed) state.
Force stationary (cm/deg)	10~1000 cm/ 0~120 deg	Setup a **forward window** to keep the presence signal suitable while a still target stands, so brief detection dropouts don't flicker presence while a person sits very still. default: 150cm/40deg * It needs to tick the enable box to turn on this function.
Max still time	0 ~ 3000	The tracking dot on the radar plot will vanish after the target remains stationary for the **configured seconds.** (Sec) * It needs to tick the enable box to turn on this function.
Adv clutter reject (sth/dth)	0~1000 sth/ 0~50 dth	Reject repetitive background motion (AC, machinery, …) so it is not reported as presence. Turn it off to report every moving target. default: 300sth / 20dth * It needs to tick the enable box to turn on this function.
Power down (sleep)	-	Power down the sensing function to save power. * It needs to tick the enable box to turn on this function.
GPIO human flag output	-	Enable the GPO to output the human flag signal. * It needs to tick the enable box to turn on this function.
Tracker state advanced mode	-	Enable the advanced tracker state mode. * It needs to tick the enable box to turn on this function.

- Configuration parameters can be saved to persistent memory; they will remain intact after a power cycle and be loaded at the next startup.

Function	Description
Factory Reset	Clicking “Factory Reset” will clear the saved NVM (Non-Volatile Memory) settings in the MCU and restore the system to its factory firmware defaults.
Save Setting	Clicking “Save Setting” will overwrite the MCU’s stored NVM configuration with the current settings. **The saved configuration remains effective even after a power outage.**
Initial	Loads the configuration parameters currently stored in the MCU’s NVM.
Clear State	Reset the presence flag, and detection restarts over; settings are still kept.

- **Measurement Quality Indicators & Troubleshooting**

Use the following parameters to assess signal health and measurement validity

Indicator / Parameter	Expected Pass Criteria	Recommended Action if abnormal Condition
Motion/Stationary Magnitude	$\text{Mag\_m} > \text{Mag floor}_{\text{motion}}$ $\text{Mag\_s} > \text{Mag floor}_{\text{stationary}}$	**1. Verify Background Noise Floor:** Clear the detection zone and observe the ambient magnitude on the GUI plot. Ensure the noise floor is well below Mag floor. **2.** If signal is weak, increase Gain setting (+3dB to +21dB), and increase Stationary Sensitivity if needed.
False Alarm / Ghost Targets	Human Flag = 0 when the room is empty.	**1. Check Detection Range Boundary:** Verify if Room range or FOV parameters exceed physical room boundaries or wall limits, catching outside motion through thin partitions. **2. Check Metallic Reflector Obstacles:** Inspect for large metal surfaces (e.g., metal cabinets, large mirrors, metallic whiteboards) in the FOV that cause multipath RF reflections. Relocate sensor or re-orient away from metal structures. **3. Enable Clutter Filtering:** Turn on Adv clutter reject or raise Mag floor to filter out background machinery/AC vibrations.

Ordering Information

Contact: Q@keywavetech.co.uk

Item	**Description**
Product	KW307 Human Presence Sensor Module
Part number	KW307
Module variant	KW307-P
Product name	KW307 Human Presence Sensor Module
Sensor type	24 GHz RF-based human presence sensor module
Interface	UART, GPO, programming interface pins
Supply options	3.6V ~ 5V supply input
Package / module type	PCB module with integrated antenna
Module dimensions	Approximately 23.6 mm × 21.1 mm
Pin header	2.54 mm pitch on KW307 module
Hardware version	KW307-P-M1
Firmware version	v2.7.6
GUI revision	vB.2
Datasheet version	v2.5
Evaluation support	PC GUI configuration and monitoring tool

For the latest updates, visit our website: www.keywavewtech.co.uk

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